

Constructors

* Constructors

```
class Student {  
    String name;  
    int age;  
    int rollNo;  
    String college;  
  
    Student() {  
        name = "Aditya"  
        :  
    }  
}
```

Student s1 = new Student();

cout (s1. name)
(s1. rollNo)

used to call constructor

we'll get these values

// setting
default values

Rules of constructors

- ① same name as class
- ① no return type (not even void)
- ① automatically called during object creation
- ① used to initialize an object
- ① it can also be overloaded

```
class Student {  
    String name  
    _____  
    _____  
    _____  
  
    Student() {  
        // empty  
    }  
}
```

Student s1 = new Student();

Constructors is mandatory

↓
default constructor (if you don't
create constructor)

for dynamic values

Parameterized Constructor

```
class Student {  
    String name;  
    int age;  
    int rollNo;  
    String college;  
}
```

```
Student (String n, int a,  
        int rn, String c)  
{  
    name = n;  
    age = a;  
    rollNo = rn;  
    college = c;  
}
```

```
Student () {  
}
```

```
Student s1 = new Student ("Aditya", 18,  
                          121, "NIT  
                          Trichy");
```

```
Student s2 = new Student ();
```

if empty constructor is not initialized.
compile error (now no error)

* this keyword → refer to current object

```
class Student {
```

```
    String name;  
    int age;  
    int rollNum;  
    String college;
```

```
    Student (String name, int age, int rollNum, String college) {
```

```
        this.name = name;
```

```
        this.age = age;
```

```
        this.rollNum = rollNum;
```

```
        this.college = college;
```

```
    }
```

```
}
```


* Chaining in constructor

↓
constructor overload → Chaining

// we'll explain this much better in code.

Q. Can we call constructor manually?

→ No

→ We can only call constructor during object initialization or using this constructor inside the object class.

Q. if memory is full in Runtime

we get Runtime exception when heap memory is full during Run time cause object is stored in runtime dynamically.